MINI PROJECT DOCUMENTATION

FairHire AI

AI-Powered Fake Job & Recruitment Scam Detection Platform

Domain: Computer Science & Engineering (Artificial Intelligence & Machine Learning)

Development Platform: Base44 AI

Live Project

https://pastel-safe-hire-scan.base44.app

Mini Project Report

1. Abstract

FairHire AI is an AI-powered web application designed to help students, fresh graduates, and job seekers identify potentially fake, fraudulent, or suspicious job advertisements. The platform analyzes job-related information and produces a risk-oriented assessment along with detected warning signs, explanations, and safety recommendations.

The project focuses on applying Artificial Intelligence and Natural Language Processing concepts to a practical recruitment-safety problem. The web application was developed using Base44 AI as the primary application-development platform. The system is intended as a decision-support and awareness tool; an AI risk assessment is not definitive proof that a job is fraudulent.

2. Introduction

Online recruitment has become an important part of modern employment and internship searches. At the same time, fake job advertisements and recruitment scams can expose applicants to financial loss, privacy risks, identity theft, and other harmful outcomes.

FairHire AI addresses this problem by providing a simple interface through which a user can submit job information for analysis. The application evaluates potential warning signs such as unrealistic salary claims, suspicious recruiter information, payment requests, urgency-based language, incomplete employer details, and questionable application processes.

3. Problem Statement

Job seekers often find it difficult to determine whether an online job advertisement is legitimate or suspicious before applying. Manual verification can be time-consuming, and inexperienced applicants may overlook important warning signs. Therefore, a web-based intelligent system is required to analyze job information, identify potential recruitment-scam indicators, generate an understandable risk assessment, and provide safety guidance.

4. Objectives

• Develop an AI-powered web platform for recruitment-scam risk assessment.

• Apply AI/NLP-based analysis to job advertisement information.

• Identify suspicious recruitment patterns and warning signs.

• Generate an interpretable risk score and risk level.

• Provide safety recommendations to users.

• Maintain a user-friendly and responsive interface.

• Demonstrate rapid AI-assisted application development using Base44 AI.

5. Proposed Solution

FairHire AI provides a centralized platform where users can enter job details and analyze them for potentially suspicious indicators. The result presents a risk-oriented assessment rather than a simple binary 'fake' or 'real' decision. This makes the output easier to understand and encourages users to independently verify important employment information.

6. Key Features

• Job advertisement analysis.

• AI/NLP-based suspicious-pattern analysis.

• Risk score and risk-level visualization.

• Explainable warning indicators.

• Safety recommendations.

• Analysis history.

• Suspicious-job reporting.

• User and administrative views where configured.

• Responsive web interface.

• Educational recruitment-safety information.

7. System Workflow

```
User
  ↓
Enter / Paste Job Information
  ↓
Text & Information Analysis
  ↓
Suspicious Pattern Detection
  ↓
Risk Assessment
  ↓
Risk Score + Risk Level
  ↓
Warning Signs + Explanations
  ↓
Safety Recommendations
```

8. AI Analysis Approach

The application is designed around an AI-assisted analysis workflow. Job information can be examined for signals associated with recruitment scams, including unrealistic compensation, requests for payment, suspicious recruiter communication, urgency or pressure tactics, incomplete company information, questionable application links, unrealistic promises, and unusual job-description patterns.

Risk Levels

Risk Level	Interpretation
Low Risk	Fewer potentially suspicious indicators detected.
Medium Risk	Some potentially concerning indicators detected.
High Risk	Multiple potentially suspicious indicators detected.

9. Main Modules

9.1 Home Module

Introduces FairHire AI, explains its purpose, and guides users toward job analysis.

9.2 Job Analysis Module

Collects relevant job information such as job title, company, location, salary, recruiter details, job description, and application information.

9.3 AI Risk Analysis Module

Analyzes the submitted information and generates a risk-oriented assessment with suspicious indicators.

9.4 Result Module

Displays the risk score, risk level, detected warning signs, explanations, and safety recommendations.

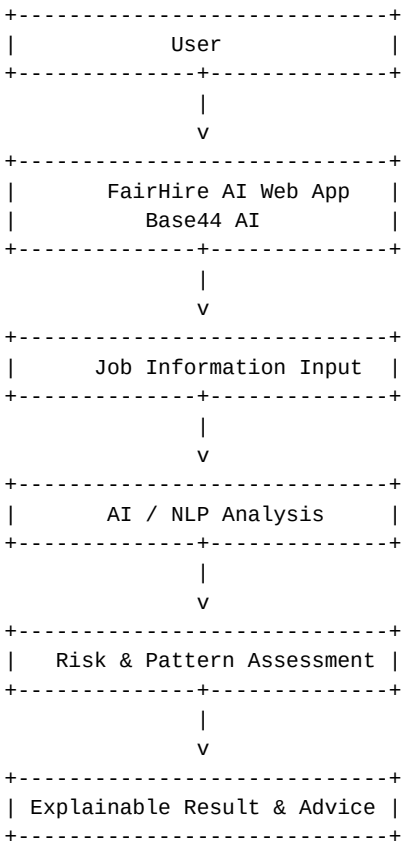
9.5 History Module

Allows users to review previously analyzed job opportunities.

9.6 Safety Information Module

Provides educational guidance about common recruitment scams and safe job-search practices.

10. System Architecture



11. Technologies Used

Category	Technology / Concept
Application Development	Base44 AI
Web	Responsive web application technologies
AI	Artificial Intelligence
NLP	Natural Language Processing / text analysis

Analytics	Risk scoring and pattern-based assessment
Version Control	Git and GitHub
Deployment	Base44 hosted web application

12. Use of Base44 AI

Base44 AI was used as the primary AI-assisted application-development platform for building the project. It supported the creation of the user interface, application pages, components, application logic, data handling, responsive design, and AI-oriented functionality. The project demonstrates how an AI-assisted development platform can be used to rapidly prototype and deploy a functional web application.

13. User Flow

• Open the FairHire AI web application.

• Navigate to the job-analysis feature.

• Enter or paste available job information.

• Submit the information for analysis.

• Review the generated risk score and risk level.

• Read the detected warning signs and explanations.

• Follow the recommended safety actions.

• Review previous analyses when required.

14. Real-World Applications

• Students searching for internships and placements.

• Fresh graduates applying for entry-level jobs.

• General job seekers using online recruitment platforms.

• College placement and career guidance cells.

• Recruitment-safety awareness programs.

• Career-support platforms.

15. Limitations

• The assessment depends on the quality and completeness of the information provided.

• An AI-generated risk score does not prove that a job is fraudulent.

• Independent company and recruiter verification may still be required.

• AI analysis can produce false positives or false negatives.

• New scam patterns may not always be detected immediately.

16. Future Enhancements

• Integration with company and domain verification APIs.

• Advanced machine-learning classification models trained on verified datasets.

• Multilingual job-advertisement analysis.

• Duplicate and near-duplicate job detection.

• Email and domain reputation analysis.

• Browser extension for job-search websites.

• Community-based reporting and verification.

• Mobile application.

• Advanced explainable-AI visualizations.

• Model-performance monitoring and evaluation.

17. Ethical Considerations

FairHire AI should be used as a risk-assessment and awareness tool. The system should not be used to make definitive accusations against individuals or organizations solely on the basis of an AI-generated score. Users should independently verify employers and employment opportunities before sharing sensitive information or making payments.

18. Expected Outcome

The expected outcome is a functional web application capable of accepting job advertisement information, analyzing potential recruitment-scam indicators, generating a risk-oriented assessment, explaining suspicious signals, and providing safety recommendations. The project also demonstrates practical use of AI-assisted software development with Base44 AI.

19. Live Project

FairHire AI: https://pastel-safe-hire-scan.base44.app

The application is deployed as a Base44-hosted web application and can be accessed through the live project link.

20. Conclusion

FairHire AI demonstrates the practical application of Artificial Intelligence and Natural Language Processing concepts to a real-world recruitment-safety problem. By combining AI-assisted analysis with a simple web interface, the platform can help users recognize potential warning signs in online job opportunities and make more informed decisions.

The project also demonstrates the value of Base44 AI as a rapid application-development platform for creating and deploying functional web applications. With further integration of verified datasets, external verification services, and advanced machine-learning models, the platform can be expanded into a more comprehensive recruitment-safety solution.

21. GitHub Repository Structure

```
FairHire-AI/
■
■■■ README.md
■■■ documentation/
```

```
■   ■■■ Mini_Project_Documentation.pdf
■
■■■ screenshots/
■   ■■■ home.png
■   ■■■ analysis.png
■   ■■■ result.png
■   ■■■ dashboard.png
■
■■■ assets/
    ■■■ logo.png
```

22. Project Information

Project Title	AI-Powered Fake Job & Recruitment Scam Detection Platform
Project Name	FairHire AI
Project Type	B.Tech Mini Project
Specialization	CSE – Artificial Intelligence & Machine Learning
Development Tool	Base44 AI
Live URL	https://pastel-safe-hire-scan.base44.app